

Day-12

DATE:21-08-25

NAME: NANCY M

[AppDbContext.cs:](#)

```
using Day13task.Models;
using Microsoft.EntityFrameworkCore;

namespace Day13task.Context
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) :
base(options)
        {

        }

        public DbSet<ChiefMinister> ChiefMinisters { get; set; }
        public DbSet<Usermembers> usermembers { get; set; }

    }
}
```

[MinisterController.cs](#)

```
using Day13task.Context;
using Day13task.Models;
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Mvc;

namespace Day13task.Controllers
{
    [ApiController]
    [Route("[controller]")]
    public class MinisterController : Controller
    {
        AppDbContext appDbContext;
```

```

public MinisterController(AppDbContext ctxt)
{
    appDbContext = ctxt;
}
[Authorize(Roles = "Admin")]
[HttpPost("AddDetails")]
public IActionResult AddMinister(ChiefMinister chiefMinister)
{
    appDbContext.ChiefMinisters.Add(chiefMinister);
    appDbContext.SaveChanges();
    return Ok(" added");
}
[Authorize(Roles = "Officer")]
[HttpGet("FetchAllMinisters")]
public IActionResult FetchAllMinisters()
{
    var allMinister = appDbContext.ChiefMinisters.ToList();
    return Ok(allMinister);
}
}
}

```

[UserController.cs](#)

```

using Day13task.Context;
using Day13task.Models;
using Day13task.Security;
using Microsoft.AspNetCore.Mvc;
using System.Security.Claims;
using System.Security.Cryptography;
using System.Text;

namespace Day13task.Controllers
{
    public record LoginRequest(string name, string Password);
    [ApiController]
    [Route("[controller]")]
    public class UserController : Controller
    {
        AppDbContext appDbContext;
    }
}

```

```

    Microsoft.Extensions.Options.IOptions<JWTOptions> jwtOptions;
    public UserController(AppDbContext ctxt,
Microsoft.Extensions.Options.IOptions<JWTOptions> _jwtOptions)
    {
        appDbContext = ctxt;
        jwtOptions = _jwtOptions;
    }
    const string ADMIN = "admin";
    const string OFFICER = "officer";
    [HttpPost("Login")]
    public IActionResult Login(LoginRequest loginReq)
    {
        var name = loginReq.name;
        var hashedPassword = GetPassHash(loginReq.Password);
        var logged_in_user = appDbContext.usermembers.Where(x => x.Name ==
name && x.Password == hashedPassword).FirstOrDefault();
        if (logged_in_user == null)
        {
            return Unauthorized("sry unauthorized");
        }
        else
        {
            var claims = new List<Claim>
            {
                new Claim(ClaimTypes.Name,loginReq.name),
                //new Claim(ClaimTypes.Role,"Admin")
            };
            if (logged_in_user.Role.Trim().ToLower().Contains(ADMIN))
            {
                claims.Add(new Claim(ClaimTypes.Role, "Admin"));
            }
            if (logged_in_user.Role.Trim().ToLower().Contains(OFFICER))
            {
                claims.Add(new Claim(ClaimTypes.Role, "Officer"));
            }

            var token = JwtService.CreateJWTToken(jwtOptions.Value, claims);
            return Ok(token);
        }
    }
}

```

```

[HttpPost("AddUser")]
public IActionResult AddUser(Usermembers user)
{
    Usermembers user_with_hashed_pwd = new Usermembers { Name =
user.Name, Email = user.Email, Id = user.Id, Password =
GetPassHash(user.Password), Role = user.Role };
    appDbContext.usermembers.Add(user_with_hashed_pwd);
    appDbContext.SaveChanges();
    //return Ok("user added successfully");
    return RedirectToAction("GetAllUsers");
}

[HttpGet("GetAllusers_v2")]
public IActionResult GetAllUsers()
{
    var allUsers = appDbContext.usermembers.Select(x => new { x.Name,
x.Email }).ToList();
    return Ok(allUsers);
}

private string GetPassHash(string raw_pass)
{
    using (SHA256 sha256Hash = SHA256.Create())
    {
        byte[] pass_converted_to_bytes = Encoding.UTF8.GetBytes(raw_pass);
        byte[] bytes = sha256Hash.ComputeHash(pass_converted_to_bytes);
        StringBuilder builder = new StringBuilder();
        for (int i = 0; i < bytes.Length; i++)
        {
            builder.Append(bytes[i].ToString("x2"));
        }
        return builder.ToString();
    }
}
}
}
}

```

ChiefMinister.cs:

```
using System.ComponentModel.DataAnnotations;
namespace Day13task.Models
{
    public class ChiefMinister
    {
        [Key]
        public int Id { get; set; }
        public string State { get; set; }
        public string Chief_Minister { get; set; }
        public string Language { get; set; }
    }
}
```

Usermember.cs:

```
using System.ComponentModel.DataAnnotations;
namespace Day13task.Models
{
    public class Usermembers
    {
        public int Id { get; set; }
        [StringLength(25), MinLength(5)]
        public string Name { get; set; }
        [StringLength(25), MinLength(5)]
        public string? Email { get; set; }
        public string Password { get; set; }
        [StringLength(25), MinLength(5)]
        public string? Role { get; set; } = "user";
    }
}
```

GET

http://localhost:5247/Minister/FetchAllMinisters

Send

Params Authorization Headers (9) Body Scripts Settings

Cookies

Auth Type

Bearer Token

Token

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results (1/1)

200 OK • 103 ms • 397 B • Save Response

JSON

Preview

Visualize

```
1 [
2   {
3     "id": 1,
4     "state": "Tamil Nadu",
5     "chief_Minister": "Stalin",
6     "language": "Tamil"
7   },
8   {
9     "id": 3,
10    "state": "Kerala".
```

POST

http://localhost:5247/Minister/AddDetails

Send

Params Authorization Headers (9) Body Scripts Settings

Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON

Beautify

```
1 {
2   "state": "Karnataka",
3   "chief_Minister": "Chandranaidu Baabu",
4   "language": "Kanadam"
```

Body Cookies Headers (4) Test Results (1/1)

200 OK • 428 ms • 148 B • Save Response

Raw

Preview

Visualize

1 added

```
SELECT TOP (1000) [Id]
, [State]
, [Chief_Minister]
, [Language]
FROM [demo_nancy].[dbo].[ChiefMinisters]
```

100 %



Results



Messages

	Id	State	Chief_Minister	Language
1	1	Tamil Nadu	Stalin	Tamil
2	3	Kerala	Pinarayi Vijayan	Malayalam
3	4	Karnataka	Chandranaidu Baabu	Kannada

SQLQuery28.sql -...YODA\nancy.m (59)) SQLQuery27.sql -...YODA\nancy.m (56)) SQLQuery26.sql -...ODA\nanc

```
SELECT TOP (1000) [Id]
, [Name]
, [Email]
, [Password]
, [Role]
FROM [demo_nancy].[dbo].[usermembers]
```

100 %

Results Messages

	Id	Name	Email	Password	Role
1	1	maria	maria@gmail.com	94aec9fbed989ece189a7e172c9cf41669050495152bc4c1...	admin
2	2	maria	maria@gmail	94aec9fbed989ece189a7e172c9cf41669050495152bc4c1...	admin
3	3	maria	maria@gmail.com	94aec9fbed989ece189a7e172c9cf41669050495152bc4c1...	admin
4	4	maria	maria@gmail.com	94aec9fbed989ece189a7e172c9cf41669050495152bc4c1...	admin
5	5	maria	maria@gmail.com	94aec9fbed989ece189a7e172c9cf41669050495152bc4c1...	admin
6	6	nimal	nimal@gmail.com	8017d27151912033277faad0effc8662e0686b3602989ca7c...	officer